

Stat 254 – Full Solution Key | Sets 22–30

Extra Problem Sets + Sample Test 3 · Step-by-step with formula sheet guidance

Key: Decision keyword Given data Formula used Trap / warning Computed d values

SET 22 — Single Mean Case I: σ known $\rightarrow Z$ | Case II: σ unknown, $n \geq 30 \rightarrow Z$ | Case III: σ unknown, $n < 30 \rightarrow T$, $df=n-1$

S22 · P1 pH Meter Bias

QUESTION (SET 22 · P1)

Suppose that an engineer is interested in testing the **bias** in a pH meter. Data are collected on a neutral substance (pH = 7.0). A sample of measurements is taken, with the data as follows:

7.00, 7.10, 6.97, 7.00, 7.03, 7.01, 7.01, 6.98, 7.08

Is there sufficient evidence to indicate that the pH meter is biased? Use $\alpha = 0.05$.

FORMULA CHOICE

"Bias" = testing if the mean deviates from a known value $\rightarrow$ **single mean t-test** (σ unknown, $n=9 < 30$). "Bias" can go either way $\rightarrow$ **two-tailed**: $H_0: \mu=7.0$, $H_1: \mu \neq 7.0$. [Sheet \$\rightarrow\$ S22: Case III \(\$\sigma\$ unknown, \$n < 30 \rightarrow T\$ \)](#)

Step 1. $n=9$, $\bar{x} = (7.00+7.10+6.97+7.00+7.03+7.01+7.01+6.98+7.08)/9 = 63.18/9 = 7.0200$

Step 2. $s = 0.0436$ (compute from deviations or calculator)

Step 3. $t = (7.0200 - 7.0) / (0.0436/\sqrt{9}) = 0.020 / 0.01453 = 1.38$, $df = 8$

Step 4. t-table: $t_{0.10,8}=1.397$, $t_{0.10,8}=1.397$. Since $|1.38| < 1.397 \rightarrow$ **p-value > 0.20**

Do not reject H_0 . No significant evidence of bias.

S22 · P2 Lubricant Container Content

QUESTION (SET 22 · P2)

Test the hypothesis that the **average content of containers of a particular lubricant is 10 liters** if the contents of a random sample of 10 containers are:

10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, 9.8 liters

Use a 0.01 level of significance and assume that the distribution of contents is normal.

FORMULA CHOICE

One sample, testing a specific μ value, σ unknown, $n=10 \rightarrow$ **single mean t-test**. "Average content = 10" $\rightarrow$ two-tailed.

[Sheet \$\rightarrow\$ S22: Case III \(\$\sigma\$ unknown, \$n < 30 \rightarrow T\$ \)](#)

Step 1. $\bar{x} = 100.6/10 = 10.06$ $s = 0.2459$

Step 2. $t = (10.06 - 10) / (0.2459/\sqrt{10}) = 0.06 / 0.07776 = 0.77$, $df = 9$

Step 3. $t_{0.10,9}=1.383$. Since $0.77 < 1.383 \rightarrow$ **p-value > 0.20**

Do not reject H_0 at $\alpha = 0.01$. Insufficient evidence the content differs from 10 L.

S22 · P3 Standardized Test Completion Time**QUESTION (SET 22 · P3)**

Past experience indicates that the time required for high school seniors to complete a standardized test is a normal random variable with a **mean of 35 minutes**. If a random sample of **20** high school seniors took an average of **33.1 minutes** to complete this test with a standard deviation of **4.3 minutes**, test the hypothesis at the 0.05 level of significance that $\mu = 35$ minutes against the alternative that $\mu < 35$ minutes.

FORMULA CHOICE

" $\mu < 35$ " → **one-tailed lower** test. $n=20$, σ unknown → **t-test**. $df = n-1 = 19$. [Sheet → S22: Case III \(\$\sigma\$ unknown, \$n < 30\$ → T\)](#)

Step 1. $t = (33.1 - 35) / (4.3/\sqrt{20}) = -1.9 / 0.9613 = -1.98$, $df = 19$

Step 2. One-tailed lower: $t_{0.025,19}=2.093$, $t_{0.05,19}=1.729$. Since $1.729 < |-1.98| < 2.093$ → **0.025 < p-value < 0.05**

Reject H_0 at $\alpha = 0.05$. Evidence that mean time is less than 35 minutes.

SET 23 — Single Proportion (z-test) Use when: yes/no outcome, 1 sample → $z = (\hat{p} - p_0)/\sqrt{(p_0q_0/n)}$ **S23 · P1** Heat Pump Installation Claim**QUESTION (SET 23 · P1)**

A builder **claims** that heat pumps are installed in **70%** of all homes being constructed today in the city of Richmond, Virginia. Would you agree with this claim if a random survey of new homes in this city showed that **8 out of 15** had heat pumps installed? Use a **0.10 level of significance**.

FORMULA CHOICE

"**Proportion**" (installed or not = yes/no) + testing a specific p_0 → **single proportion z-test**. "**Agree with claim**" = two-tailed (could be too high or too low). [Sheet → Single Proportion card → z_obs row](#)

Step 1. $\hat{p} = 8/15 = 0.5333$ $p_0 = 0.70$, $q_0 = 0.30$

Step 2. $z = (0.5333 - 0.70) / \sqrt{(0.70 \times 0.30/15)} = -0.1667 / 0.1183 = -1.41$

Step 3. Two-tailed: $p\text{-value} = 2 \times P(Z < -1.41) = 2 \times 0.0793$ **p-value = 0.1586**

$p\text{-value} = 0.1586 > \alpha = 0.10$ → Do not reject H_0 . No significant disagreement with the 70% claim.

S23 · P2 Lasagna Preference**QUESTION (SET 23 · P2)**

A marketing expert for a pasta-making company believes that **40%** of pasta lovers prefer lasagna. If **9 out of 20** pasta lovers choose lasagna over other pastas, what can be concluded about the expert's claim? Use $\alpha = 0.05$.

FORMULA CHOICE

Same setup as P1 — proportion test, p_0 given → **single proportion z-test**, two-tailed.

[Sheet → Single Proportion card → z_obs row](#)

Step 1. $\hat{p} = 9/20 = 0.45$ $p_0 = 0.40$, $q_0 = 0.60$

Step 2. $z = (0.45 - 0.40) / \sqrt{(0.40 \times 0.60/20)} = 0.05 / 0.1095 = 0.46$

Step 3. Two-tailed: $p\text{-value} = 2 \times P(Z > 0.46) = 2 \times 0.3228$ **p-value = 0.6456**

$p\text{-value} = 0.65 >> \alpha = 0.05$ → Do not reject H_0 . Data is consistent with the 40% claim.

SETS 24–25 — Two Independent Means Case I: σ known $\rightarrow Z$ | Case II: large n ($n_1, n_2 \geq 30$) $\rightarrow Z$ | Case III: small n , ratio $\leq 1.4 \rightarrow$ Pooled T | Case IV: small n , ratio $> 1.4 \rightarrow$ Unpooled T

S24 · P1 Two Populations, σ Known

QUESTION (SET 24 · P1)

A random sample of size $n_1 = 25$, taken from a normal population with a standard deviation $\sigma_1 = 5.2$, has a mean $\bar{x}_1 = 81$. A second random sample of size $n_2 = 36$, taken from a different normal population with a standard deviation $\sigma_2 = 3.4$, has a mean $\bar{x}_2 = 76$. Test the hypothesis at the 0.06 level of significance that $\mu_1 = \mu_2$ against the alternative that $\mu_1 \neq \mu_2$.

 FORMULA CHOICE

Two separate groups + σ (population s.d.) given $\rightarrow$ use **z-test** (not t). Two-tailed since " $\neq$ ".

[Sheet \$\rightarrow\$ S24/25: Case I \(\$\sigma\$ known \$\rightarrow Z\$ \)](#)

Step 1. $ese = \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)} = \sqrt{(27.04/25 + 11.56/36)} = \sqrt{(1.0816 + 0.3211)} = \sqrt{1.4027} = 1.1844$

Step 2. $z = (81 - 76) / 1.1844 = 5 / 1.1844 = 4.22$

Step 3. Two-tailed p-value = $2 \times P(Z > 4.22) \approx$ **≈ 0 (reject H_0)**

Strong evidence that $\mu_1 \neq \mu_2$.

S24 · P2 Abrasive Wear of Two Materials (Pooled t)

QUESTION (SET 24 · P2)

An experiment was performed to compare the **abrasive wear** of two different laminated materials. **12** pieces of material 1 were tested (average wear = **85** units, $s_1 = 4$) and **10** pieces of material 2 (average = **81**, $s_2 = 5$). Can we conclude at the **0.05 level of significance** that the abrasive wear of material 1 exceeds that of material 2 by **more than 2 units**?

 FORMULA CHOICE

Two groups, σ unknown, $s_1=4$ and $s_2=5$ are **close** (ratio=1.25 $\leq$ 1.4) $\rightarrow$ **Pooled t-test**, $df=n_1+n_2-2=20$. One-tailed upper ($H_1: \mu_1 - \mu_2 > 2$). **Note: $\Delta_0 = 2$, not 0!** [Sheet \$\rightarrow\$ S24/25: Case III \(Pooled T, small \$n\$, ratio \$\leq\$ 1.4\)](#)

Step 1. $s_p^2 = [11(16) + 9(25)] / 20 = [176 + 225] / 20 = 401 / 20 = 20.05$ $s_p = 4.478$

Step 2. $ese = 4.478 \times \sqrt{(1/12 + 1/10)} = 4.478 \times 0.4282 = 1.917$

Step 3. $t = [(85 - 81) - 2] / 1.917 = 2 / 1.917 = 1.04$, $df=20$

Step 4. One-tailed: $t_{0.10, 20} = 1.325$. Since $1.04 < 1.325 \rightarrow$ **p-value > 0.10**

Do not reject H_0 . Insufficient evidence that wear of material 1 exceeds material 2 by more than 2 units.

S25 · P3 Thread Tensile Strength (Manufacturer's Claim)**QUESTION (SET 25 · P3)**

A manufacturer **claims** that the average tensile strength of thread A **exceeds** that of thread B by **at least 12 kg**. To test this claim, **50 pieces of each** were tested. Thread A: $\bar{x}_1 = 86.7$ kg, $s_1 = 6.28$. Thread B: $\bar{x}_2 = 77.8$ kg, $s_2 = 5.61$. Test the hypothesis $H_1: \mu_1 - \mu_2 < 12$ at $\alpha = 0.05$.

FORMULA CHOICE

Both $n=50$ (large) → use **z-test**. "**Exceeds by at least 12**" means $\Delta_0=12$. H_1 is one-tailed lower. **Trap: $\Delta_0 = 12$, not 0. Subtract 12 in numerator.** [Sheet → S24/25: Case II \(large n, \$n_1=n_2=50 \geq 30 \rightarrow Z\$ \)](#)

Step 1. $s_{\text{se}} = \sqrt{(6.28^2/50 + 5.61^2/50)} = \sqrt{(0.7892 + 0.6296)} = \sqrt{1.4188} = 1.1912$

Step 2. $z = [(86.7 - 77.8) - 12] / 1.1912 = (8.9 - 12) / 1.1912 = -3.1 / 1.1912 = -2.60$

Step 3. One-tailed lower: $p\text{-value} = P(Z < -2.60) = 1 - 0.9953$ **p-value = 0.0047**

Reject H_0 . Evidence that the difference is less than 12 kg — manufacturer's claim is not supported.

S25 · P4 Tire Brand Wear (Pooled t)**QUESTION (SET 25 · P4)**

Engineers are trying to decide between two tire brands. **12 tires of each brand** were run until they wore out. Results: Brand A — $\bar{x}_1 = 37,900$ km, $s_1 = 4,500$. Brand B — $\bar{x}_2 = 39,800$ km, $s_2 = 6,900$. Is there reason to believe there is **no difference** in average wear? Assume normal populations with **equal variances**.

FORMULA CHOICE

Two groups, equal variances stated → **Pooled t-test**. "**No difference**" → two-tailed, $\Delta_0=0$.

[Sheet → S24/25: Case III \(Pooled T, small n, ratio \$\leq 1.4\$ \)](#)

Step 1. $s_p^2 = [11(4500^2) + 11(6900^2)] / 22 = [222,750,000 + 523,710,000] / 22 = 33,929,545$ $s_p = 5825$

Step 2. $t = (37900 - 39800) / (5825 \times \sqrt{2/12}) = -1900 / (5825 \times 0.4082) = -1900 / 2378 = -0.80$, $df=22$

Step 3. $t_{0.10,22}=1.321$. $|-0.80| < 1.321 \rightarrow$ **p-value > 0.20**

Do not reject H_0 . No significant difference in tire wear.

SET 26 — Paired / Matched-Pair t-test Case I: $n \geq 30 \rightarrow Z$ | Case II: $n < 30$, differences normal $\rightarrow$ Paired T,
 $df=n-1$ · $d = x_1 - x_2$, then $t = \bar{d} / (s_d / \sqrt{n})$

S26 · P1 Radial vs Belted Tires — Fuel Economy
QUESTION (SET 26 · P1)

A taxi company manager is trying to decide whether the use of **radial tires** instead of regular belted tires **improves fuel economy**. **Twelve cars** were equipped with radial tires and driven over a prescribed test course. Without changing drivers, the same cars were then equipped with belted tires and driven again. Gasoline consumption (km/liter) is recorded below. Can we conclude that radial tires give better fuel economy? Assume populations are normally distributed. Use $\alpha = 0.05$.

FORMULA CHOICE

"Same cars" measured under both treatments = matched pairs → **Paired t-test**. **Never use two-sample t here — the observations are linked.** **"Better"** → one-tailed upper: $H_1: \mu_D > 0$ ($d = \text{radial} - \text{belted}$). $n=12 < 30$.

Sheet → S26: Case II ($n < 30$, Paired T, $df=n-1$)

Car	1	2	3	4	5	6	7	8	9	10	11	12
Radial	4.2	4.7	6.6	7.0	6.7	4.5	5.7	6.0	7.4	4.9	6.1	5.2
Belted	4.1	4.9	6.2	6.9	6.8	4.4	5.7	5.8	6.9	4.7	6.0	4.9
d = R-B	0.1	-0.2	0.4	0.1	-0.1	0.1	0.0	0.2	0.5	0.2	0.1	0.3

Step 1. $\Sigma d = 1.7$ $\bar{d} = 1.7/12 = 0.1417$

Step 2. $\Sigma d^2 = 0.01+0.04+0.16+0.01+0.01+0.01+0+0.04+0.25+0.04+0.01+0.09 = 0.67$

$s_{d^2} = [0.67 - 12(0.1417)^2]/11 = [0.67 - 0.2408]/11 = 0.4292/11 = 0.0390$ $s_d = 0.1975$

Step 3. $t = 0.1417 / (0.1975/\sqrt{12}) = 0.1417 / 0.05701 = 2.49$, $df=11$

Step 4. One-tailed: $t_{0.01,11}=2.718$, $t_{0.025,11}=2.201$. Since $2.201 < 2.49 < 2.718 \rightarrow$ **$0.01 < p\text{-value} < 0.025$**

Reject H_0 at $\alpha=0.05$. Radial tires give significantly better fuel economy.

S26 · P2 Chemical vs X-Ray Iron Content Analysis
QUESTION (SET 26 · P2)

Five samples of a ferrous-type substance were used to determine if there is a difference between a **laboratory chemical analysis** and an **X-ray fluorescence analysis** of the iron content. Each sample was split into two subsamples and both types of analysis were applied. Data (coded iron content) shown below. Assuming normal populations, test at the **0.05 level of significance** whether the two methods give, on average, the **same result**.

FORMULA CHOICE

"Same samples" split and analyzed by both methods → **Paired t-test**. **"Same result"** → two-tailed: $H_0: \mu_D=0$, $H_1: \mu_D \neq 0$. $n=5 < 30$. [Sheet → S26: Case II \(n<30, Paired T, df=n-1\)](#)

Sample	1	2	3	4	5
X-ray	2.0	2.0	2.3	2.1	2.4
Chemical	2.2	1.9	2.5	2.3	2.4
d = C-X	0.2	-0.1	0.2	0.2	0.0

Step 1. $\bar{d} = 0.5/5 = 0.10$

Step 2. $\Sigma d^2 = 0.04+0.01+0.04+0.04+0 = 0.13$ $s_d^2 = [0.13-5(0.01)]/4 = 0.08/4 = 0.02$ $s_d = 0.1414$

Step 3. $t = 0.10 / (0.1414/\sqrt{5}) = 0.10 / 0.06325 = 1.58$, $df=4$

Step 4. Two-tailed: $t_{0.10,4}=2.132$, $t_{0.15,4} \approx 1.533$. Since $1.533 < 1.58 < 2.132 \rightarrow$ **0.10 < p-value < 0.20**

Do not reject H_0 at $\alpha=0.05$. No significant difference between the two analysis methods.

SET 27 — Two Proportions (z-test) H-test: use pooled $\hat{p}_p = (x_1+x_2)/(n_1+n_2)$ | CI: use unpooled $\hat{p}_1 = x_1/n_1 + \hat{p}_2 \hat{q}_2/n_2$
S27 · P1 Nuclear Power Plant — Urban vs Suburban
QUESTION (SET 27 · P1)

In a study to estimate the proportion of residents who favor the construction of a nuclear power plant, it is found that **63 of 100 urban residents** favor the construction while only **59 of 125 suburban residents** are in favor. Is there a **significant difference** between the proportions of urban and suburban residents who favor construction? Find the p-value.

FORMULA CHOICE

Two separate groups + proportion outcome (favor/oppose) → **two proportions z-test**. **"Difference"** → two-tailed. Must use **pooled $\hat{p}_p$** for the hypothesis test. [Sheet → Two Proportions card → z_obs row, pooled \$\hat{p}_p\$](#)

Step 1. $\hat{p}_1 = 63/100 = 0.63$ $\hat{p}_2 = 59/125 = 0.472$

Step 2. $\hat{p}_p = (63+59)/(100+125) = 122/225 = 0.5422$

Step 3. $z = (0.63-0.472) / \sqrt{(0.5422 \times 0.4578 \times (1/100+1/125))} = 0.158 / \sqrt{(0.2483 \times 0.018)} = 0.158/0.0669 = 2.36$

Step 4. Two-tailed: p-value = $2 \times P(Z > 2.36)$ **p-value = 0.0164**

Reject H_0 at $\alpha=0.05$. Significant difference between urban and suburban support.

S27 · P2 Chemical Plant — Town vs County Voters**QUESTION (SET 27 · P2)**

A vote is taken on whether a proposed chemical plant should be constructed. A poll shows that **120 of 200 town voters** favor the proposal and **240 of 500 county residents** favor it. Would you agree that the proportion of town voters favoring the proposal is **higher than** the proportion of county voters? Find the p-value.

FORMULA CHOICE

Two groups, proportion outcome → **two proportions z-test**. **"Higher than"** → one-tailed upper: $H_1: p_1 > p_2$.

[Sheet](#) → [Two Proportions card](#) → [z_obs row, pooled \$\hat{p}_p\$](#)

Step 1. $\hat{p}_1 = 120/200 = 0.60$ $\hat{p}_2 = 240/500 = 0.48$

Step 2. $\hat{p}_p = (120+240)/(200+500) = 360/700 = 0.5143$

Step 3. $z = (0.60 - 0.48) / \sqrt{0.5143 \times 0.4857 \times (1/200 + 1/500)} = 0.12 / \sqrt{0.2498 \times 0.007} = 0.12 / 0.04182 = 2.87$

Step 4. One-tailed upper: p-value = $P(Z > 2.87)$ **p-value = 0.0018**

Reject H_0 . Strong evidence that a higher proportion of town voters favor the plant.

SETS 28–30 — Linear Regression Always compute S_{xx} , S_{yy} , S_{xy} first → $b_1 = S_{xy}/S_{xx}$, $b_0 = \bar{y} - b_1\bar{x}$, $r = S_{xy}/\sqrt{(S_{xx} \cdot S_{yy})}$, $SSE = S_{yy} - b_1 \cdot S_{xy}$

S28 · P1 Flame-Retardant Fabric — Stiffness vs Thickness**QUESTION (SET 28 · P1)**

Physical properties of **6 flame-retardant fabric samples** were investigated. Stiffness x (mg-cm): **7.98, 24.52, 12.47, 6.92, 24.11, 35.71**. Thickness y (mm): **0.28, 0.65, 0.32, 0.27, 0.81, 0.57**.

(a) Compute the covariance, correlation coefficient r , and R^2 . (b) Estimate slope b_1 and intercept b_0 . (c) Find the residual at $x = 6.92$. (d) Predicted thickness when $x = 20$. (e) Test for linear relationship at $\alpha = 0.05$.

FORMULA CHOICE

Quantitative x predicting quantitative $y \rightarrow$ **Linear Regression**. Start with the three Sums (S_{xx} , S_{yy} , S_{xy}) — everything else follows. [Sheet](#) [→ Regression card](#) [→ Sums block first](#)

(a) Covariance, r , R^2

$$n=6, \Sigma x=111.71, \Sigma y=2.90, \Sigma x^2=2895.83, \Sigma y^2=1.5317, \Sigma xy=58.835$$

$$S_{xx} = 2895.83 - (111.71)^2/6 = 2895.83 - 2079.28 = 644.94 \text{ (use calculator)}$$

Actually from Python: $S_{xx}=644.94$, $S_{yy}=0.2555$, $S_{xy}=9.9218$

$$\text{Cov} = S_{xy}/(n-1) = 9.9218/5 = 1.984$$

$$r = S_{xy}/\sqrt{(S_{xx} \cdot S_{yy})} = 9.9218/\sqrt{(644.94 \times 0.2555)} = 9.9218/12.843 \quad \mathbf{r = 0.773}$$

$$R^2 = r^2 = 0.773^2 \quad \mathbf{R^2 = 0.597} \quad (59.7\% \text{ of variation in thickness explained by stiffness})$$

(b) Slope b_1 and Intercept b_0

$$b_1 = S_{xy}/S_{xx} = 9.9218/644.94 \quad \mathbf{b_1 = 0.0154}$$

$$\bar{x} = 111.71/6 = 18.618 \quad \bar{y} = 2.90/6 = 0.4833$$

$$b_0 = 0.4833 - 0.0154(18.618) = 0.4833 - 0.2867 \quad \mathbf{b_0 = 0.1969}$$

Regression line: $\hat{y} = 0.1969 + 0.0154x$

(c) Residual at $x = 6.92$

$$\hat{y} = 0.1969 + 0.0154(6.92) = 0.1969 + 0.1066 = 0.3035$$

$$e = y - \hat{y} = 0.27 - 0.3035 \quad \mathbf{e = -0.03}$$

(d) Predicted at $x = 20$

$$\hat{y} = 0.1969 + 0.0154(20) = 0.1969 + 0.308 \quad \mathbf{\hat{y} = 0.50 \text{ mm}}$$

(e) Test linear relationship — slope β_1 and correlation ρ ($\alpha=0.05$)

$$t = r\sqrt{(n-2)/(1-r^2)} = 0.773 \times \sqrt{4/(1-0.5973)} = 0.773 \times \sqrt{4/0.4027} = 0.773 \times 3.151 = 2.44, df=4$$

$t_{0.025,4}=2.776$. Since $2.44 < 2.776 \rightarrow$ **p-value > 0.05** (borderline — not significant at $\alpha=0.05$)

S29 · P2 Plasma Etching — Chlorine Flow vs Etch Rate**QUESTION (SET 29 · P2)**

Plasma etching is essential to fine-line pattern transfer in semiconductor manufacturing. Data for chlorine flow x (SCCM) and etch rate y (100 Å/min):

x : 1.5, 1.5, 2.0, 2.5, 2.5, 3.0, 3.5, 3.5, 4.0

y : 23.0, 24.5, 25.0, 30.0, 33.5, 40.0, 40.5, 47.0, 49.0

(a) Covariance, r , R^2 . (b) Slope b_1 and intercept b_0 . (c) Residual at $x = 4.0$. (d) Predicted etch rate at $x = 2.7$. (e) Compute SSE. (f) Test for linear relationship at $\alpha = 0.05$.

FORMULA CHOICE

Predicting etch rate from chlorine flow $\rightarrow$ **Linear Regression**. $n=9$.

[Sheet](#) $\rightarrow$ [Regression card](#)

(a) Covariance, r , R^2

$S_{xx}=6.50$, $S_{yy}=776.06$, $S_{xy}=68.917$ (computed via calculator)

$Cov = 68.917/8 = 8.615$

Cov = 8.615

$r = 68.917/\sqrt{(6.50 \times 776.06)} = 68.917/71.03$

$r = 0.9703$

$R^2 = 0.9703^2$

$R^2 = 0.9415$

(94.1% of variation explained)

(b) Slope and Intercept

$b_1 = 68.917/6.50$

$b_1 = 10.603$

$\bar{x} = 24/9 = 2.667$ $\bar{y} = 312.5/9 = 34.722$

$b_0 = 34.722 - 10.603(2.667) = 34.722 - 28.277$

$b_0 = 6.449$

(c) Residual at $x = 4.0$

$\hat{y} = 6.449 + 10.603(4.0) = 6.449 + 42.410 = 48.859$

$e = 49.0 - 48.859$

$e = 0.14$

(d) Predicted at $x = 2.7$

$\hat{y} = 6.449 + 10.603(2.7) = 6.449 + 28.628$

$\hat{y} = 35.08$ (100 Å/min)

(e) SSE

$SSE = S_{yy} - b_1 \cdot S_{xy} = 776.056 - 10.603(68.917) = 776.056 - 730.694$

SSE = 45.36

(f) Test linear relationship ($\alpha=0.05$)

$t = 0.9703 \times \sqrt{7/(1-0.9415)} = 0.9703 \times \sqrt{7/0.0585} = 0.9703 \times 10.944 = 10.62$, $df=7$

$t_{0.025,7}=2.365$. Since $10.62 \gg 2.365 \rightarrow$

p-value ≈ 0

Reject H_0 . Strong linear relationship between chlorine flow and etch rate.

S30 · P3 Maternal Age vs Baby Birth Weight**QUESTION (SET 30 · P3)**

Medical researchers have noted that adolescent females are more likely to deliver low-birth-weight babies than adult females. Data for **maternal age x** and **birth weight y (g)** from 10 observations:

x (age): **15, 17, 18, 15, 16, 19, 17, 16, 18, 19**

y (g): **2289, 3393, 3271, 2648, 2897, 3327, 2970, 2535, 3138, 3573**

(a) Covariance, r , R^2 . (b) Slope b_1 and intercept b_0 . (c) Residual at (15, 2289). (d) Predicted weight at $x = 18.5$. (e) SSE. (f) Test for linear relationship at $\alpha = 0.05$.

FORMULA CHOICE

Predicting birth weight from age → **Linear Regression**. $n=10$. [Sheet → Regression card](#)

(a) Covariance, r , R^2

$$\Sigma x = 170, \Sigma y = 30041, \Sigma x^2 = 2910, \Sigma y^2 = 91,785,351, \Sigma xy = 515,600$$

$$S_{xx} = 2910 - 170^2/10 = 2910 - 2890 = 20.00$$

$$S_{yy} = 91,785,351 - 30041^2/10 = 91,785,351 - 90,246,168 = 1,539,183$$

$$S_{xy} = 515,600 - 170 \times 30041/10 = 515,600 - 510,697 = 4,903$$

$$\text{Cov} = 4903/9 = 544.78 \quad \text{Cov} = 544.78$$

$$r = 4903/\sqrt{(20 \times 1,539,183)} = 4903/5549.7 \quad r = 0.884$$

$$R^2 = 0.884^2 \quad R^2 = 0.781 \quad (78.1\% \text{ of variation in birth weight explained by maternal age})$$

(b) Slope and Intercept

$$b_1 = 4903/20 \quad b_1 = 245.15$$

$$\bar{x} = 170/10 = 17.0 \quad \bar{y} = 30041/10 = 3004.1$$

$$b_0 = 3004.1 - 245.15(17) = 3004.1 - 4167.55 \quad b_0 = -1163.45$$

(c) Residual at data point (15, 2289)

$$\hat{y} = -1163.45 + 245.15(15) = -1163.45 + 3677.25 = 2513.80$$

$$e = 2289 - 2513.80 \quad e = -224.80$$

(d) Predicted at $x = 18.5$

$$\hat{y} = -1163.45 + 245.15(18.5) = -1163.45 + 4535.275 \quad \hat{y} = 3371.83 \text{ g}$$

(e) SSE

$$\text{SSE} = S_{yy} - b_1 \cdot S_{xy} = 1,539,183 - 245.15(4903) = 1,539,183 - 1,201,970 \quad \text{SSE} = 337,213$$

(f) Test linear relationship ($\alpha=0.05$)

$$t = 0.884/\sqrt{[8/(1-0.781)]} = 0.884/\sqrt{(8/0.219)} = 0.884 \times 6.042 = 5.34, df=8$$

$$t_{0.025,8}=2.306. \text{ Since } 5.34 >> 2.306 \rightarrow p\text{-value} < 0.001$$

Reject H_0 . Strong linear relationship between maternal age and birth weight.

SAMPLE TEST 3 — Full Solutions Questions 1–10 with formula sheet guidance

T3 · Q1 CI for Single Proportion

SAMPLE TEST 3 · QUESTION 1

We are interested in p , the proportion of all people who are currently happy with their cell phone plans. In a small study done in 2012, it was found that in a sample of **150 people**, there were **90** who were happy with their cell phone plans. **Construct a 95% confidence interval for p .**

CHOICE

CI + **proportion** + 1 sample → **Single Proportion CI**. Use $ese = \sqrt{(\hat{p}\hat{q}/n)}$, **not** $\sqrt{(p_0q_0/n)}$ for CIs.

[Sheet → Single Proportion → CI row](#)

- $\hat{p} = 90/150 = 0.6$ $ese = \sqrt{(0.24/150)} = 0.04$ $z^* = 1.96$
- $0.6 \pm 1.96(0.04) = 0.6 \pm 0.0784$ **(0.5216, 0.6784)**

T3 · Q2 Sample Size — Proportion

SAMPLE TEST 3 · QUESTION 2

(Refers to Q1 setup.) **How large should our sample be** to estimate p within a **4% margin of error** at a **95% confidence level**?

CHOICE

"How large" + **"margin of error"** → sample size formula. Use $\hat{p} = 0.6$ from Q1 (given prior estimate).

[Sheet → Single Proportion → Sample size row](#)

- $n = (1.96/0.04)^2 \times 0.6 \times 0.4 = (49)^2 \times 0.24 = 576.24 \rightarrow \text{round up}$ **$n = 577$**

If no prior estimate: use $\hat{p} = 0.5 \rightarrow n = 601$.

T3 · Q3 Two Proportions — Contaminated Broilers

SAMPLE TEST 3 · QUESTION 3

Recent incidents of food contamination have caused great concern among consumers. A report showed that **35 of 80** randomly selected **Perdue** brand broilers tested positively for campylobacter or salmonella, whereas **66 of 160 Tyson** brand broilers tested positive. Does it appear that the true proportion of contaminated broilers **differs** between the two brands? **Find the p-value.**

CHOICE

Two brands + **proportion** + **"differ"** (two-tailed) → **two proportions z-test** with pooled $\hat{p}_p$.

[Sheet → Two Proportions card](#)

- $\hat{p}_1 = 0.4375$, $\hat{p}_2 = 0.4125$, $\hat{p}_p = 101/240 = 0.4208$
- $z = 0.025 / \sqrt{(0.4208 \times 0.5792 \times 0.01875)} = 0.025 / 0.0676 = 0.370$
- $p\text{-value} = 2 \times P(Z > 0.370) = 2 \times 0.3557$ **$p\text{-value} = 0.7114$**

No evidence proportions differ.

T3 · Q4 Paired t-test — Reading Course

SAMPLE TEST 3 · QUESTION 4

10 people were given a one-day crash course in reading. Each adult was given a reading test **before** the course and another **after** the course. Results shown in the table below. For this **matched pairs experiment**, what is the observed value of the test statistic in order to test the hypothesis that **taking the course improves test scores**?

CHOICE

"Same adults" + "before/after" → **paired t-test**. "Improves" → one-tailed upper. $n=10 < 30$.

Sheet → S26: Case II ($n < 30$, Paired T, $df = n - 1$)

Adult	1	2	3	4	5	6	7	8	9	10
After	48	42	43	34	50	30	43	38	41	38
Before	31	34	18	30	44	28	34	33	27	32
d=A-B	17	8	25	4	6	2	9	5	14	6

- $\bar{d} = 96/10 = 9.6$ $\sum d^2 = 1372$ $s_d^2 = [1372 - 10(9.6)^2] / 9 = 450.4 / 9 = 50.04$ $s_d = 7.074$
- $t = 9.6 / (7.074 / \sqrt{10}) = 9.6 / 2.237$ **$t = 4.29$, $df = 9$**

Reject H_0 . Course significantly improves reading scores.

T3 · Q5 ESE — Two Independent Means (Pooled)

SAMPLE TEST 3 · QUESTION 5

Summary statistics computed for two independent samples are: $n_1 = 25$, $\bar{x}_1 = 7.30$, $s_1 = 1.05$, $n_2 = 25$, $\bar{x}_2 = 6.80$, and $s_2 = 1.20$. The **estimated standard error (ese)** of the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is equal to (assume normality)?

CHOICE

$s_1 = 1.05$, $s_2 = 1.20$ are close (ratio = $1.14 \leq 1.4$), small $n \rightarrow$ **Pooled T**. Sheet → S24/25: Case III (Pooled T, small n , ratio ≤ 1.4)

- $s_p^2 = [24(1.1025) + 24(1.44)] / 48 = 61.02 / 48 = 1.271$ $s_p = 1.128$
- $ese = 1.128 \times \sqrt{(2/25)} = 1.128 \times 0.2828$ **$ese = 0.32$**

T3 · Q6–7 Regression — Inventory Turnover

SAMPLE TEST 3 · QUESTIONS 6–7

A manager wishes to estimate the relationship between the value of inventory items and their annual turnover rate in order to **predict turnover rate**. Data from 6 items:

Value of item (x): **1, 1, 2, 2, 3, 4** | Turnover rate (y): **2, 2, 3, 5, 6, 7**

Given: $\sum x = 13$, $\sum y = 25$, $\sum x^2 = 35$, $\sum y^2 = 127$.

Q6: Find the **slope** of the regression line. **Q7:** Find the **predicted number of turnovers** per year for an item valued at \$1.50.

CHOICE

Predict y from x → **Regression**. Compute $\sum xy$ first (not given). Sheet → Regression → Sums block, then b_1 , b_0

- $\sum xy = 2+2+6+10+18+28 = 66$ $s_{xx} = 35 - 169/6 = 6.833$ $s_{xy} = 66 - 325/6 = 11.833$
 $b_1 = 11.833 / 6.833$ **$b_1 = 1.73$**
- $\bar{x} = 2.167$, $\bar{y} = 4.167$ $b_0 = 4.167 - 1.732(2.167) = 0.415$ $\hat{y} = 0.415 + 1.732(1.5)$ **$\hat{y} = 3.01$**

T3 · Q8–9 Two Large Independent Means — Restaurant
SAMPLE TEST 3 · QUESTIONS 8–9

A restaurant owner wants to determine whether customers who sit **by the window** and customers who sit in a **central location** differ in the average length of time they stay. Window sample: $n_1 = 50$, $\bar{x}_1 = 65$ min, $s_1 = 15$ min. Central sample: $n_2 = 45$, $\bar{x}_2 = 55$ min, $s_2 = 10$ min. Conduct a hypothesis test at $\alpha = 0.05$.

Q8: Find the **p-value**. **Q9:** Find a **95% confidence interval** for the difference in stay time.

CHOICE

$n_1=50, n_2=45$ both $\geq 30 \rightarrow$ **z-test**. **"Differ"**=two-tailed. [Sheet \$\rightarrow\$ S24/25: Case II \(large \$n\$, \$n_1 \geq 30\$ & \$n_2 \geq 30 \rightarrow Z\$ \)](#)

Q8. $ese = \sqrt{(225/50 + 100/45)} = \sqrt{6.722} = 2.593$ $z = 10/2.593$ **$z = 3.86, p \approx 0$**

Q9. $10 \pm 1.96(2.593) = 10 \pm 5.082$ **$(4.918, 15.082)$**

T3 · Q10 Two Small Samples — Salaries (Full Short Answer)
SAMPLE TEST 3 · QUESTION 10 (SHORT ANSWER)

Two random samples of the annual salaries of **females** and **males** were obtained. Females: $n_1 = 5$, $\bar{x}_1 = 43.9$, $s_1 = 5.4$. Males: $n_2 = 7$, $\bar{x}_2 = 42.8$, $s_2 = 2.7$. Let μ_1 and μ_2 denote the true mean salaries. The populations are **normally distributed**.

(a) Find the estimated value of $\mu_1 - \mu_2$ and the ese. (b) Construct a **98% CI** for $\mu_1 - \mu_2$. (c) State H_0 and H_1 to test if females earn **more** on average. (d) Find t_{obs} . (e) Find the p-value. (f) Draw the conclusion.

CHOICE

$s_1=5.4$ is **twice** $s_2=2.7 \rightarrow$ ratio=2.0>1.4 $\rightarrow$ **Unpooled T** (Welch). $n < 30 \rightarrow$ t-test.

[Sheet \$\rightarrow\$ S24/25: Case IV \(Unpooled T, small \$n\$, ratio>1.4\)](#)

(a) $\bar{x}_1 - \bar{x}_2 = 1.1$ $ese = \sqrt{(29.16/5 + 7.29/7)} = \sqrt{6.873}$ **$ese = 2.62$**

(b) $1.1 \pm 3.365(2.62) = 1.1 \pm 8.82$ **$(-7.72, 9.92)$** ($t^* \approx 3.365$ for 98% CI, small df)

(c) **"Higher than"** $\rightarrow$ one-tailed upper: $H_0: \mu_1 - \mu_2 = 0$ $H_1: \mu_1 - \mu_2 > 0$

(d) $t = 1.1/2.62$ **$t = 0.4196$**

(e) $t = 0.42$, small df $\rightarrow$ **$p\text{-value} > 0.10$**

(f) $p\text{-value} > 0.10 > \alpha = 0.05 \rightarrow$ Do not reject H_0 . No evidence females earn more than males.

TEST 3A — 2025 (April) Questions 1–10 · Full solutions with formula sheet guidance

2025 · Q1 Single Proportion — Oscilloscope Ownership

TEST 3A 2025 · QUESTIONS 1 & 2 REFER TO THIS SCENARIO

In a survey of **500 electrical engineering students** at a university, it is discovered that **320** of them own a personal oscilloscope. Let p represent the proportion of electrical engineering students who own a personal oscilloscope.

Q1: Compute the **p-value** for the hypothesis test $H_0: p = 0.6$ vs $H_1: p > 0.6$.

 FORMULA CHOICE

"proportion" + 1 sample + specific p_0 → **single proportion z-test**. $H_1: p > 0.6$ → **one-tailed upper**.

[Sheet](#) → [Single Proportion card](#) → [z_obs row](#)

Step 1. $\hat{p} = 320/500 = 0.64$ $p_0 = 0.60$, $q_0 = 0.40$

Step 2. $z = (0.64 - 0.60) / \sqrt{(0.60 \times 0.40 / 500)} = 0.04 / 0.02191 = 1.83$

Step 3. One-tailed upper: $p\text{-value} = P(Z > 1.83) = 1 - 0.9664$ **p-value = 0.0336**

Since $p\text{-value} = 0.0336 < 0.05$, there is evidence that more than 60% of students own an oscilloscope.

2025 · Q2 Sample Size — Proportion (No Pilot)

TEST 3A 2025 · Q2

Do **not** use the above data as a pilot study. Determine the number of students needed to estimate p within **0.03** with **94% confidence**.

 FORMULA CHOICE

"How many" + **"margin of error"** + no prior estimate → use $\hat{p} = 0.5$ (worst case). 94% CI → $\alpha/2 = 0.03$ → $z_{\{0.03\}}$.

[Sheet](#) → [Single Proportion](#) → [Sample size row](#)

Step 1. 94% confidence → $\alpha = 0.06$ → $\alpha/2 = 0.03$ → $z_{\{0.03\}} = 1.88$

Step 2. $n = z_{\{\alpha/2\}}^2 / (4E^2) = 1.88^2 / (4 \times 0.03^2) = 3.5344 / 0.0036 = 981.78$ → round up **n = 982**

Formula uses $\hat{p} = 0.5$ giving $n = z^2 / (4E^2)$. If prior estimate existed, $n = z^2 \cdot \hat{p} \hat{q} / E^2$.

2025 · Q3 Single Mean — Light Bulb Lifespan (Hypotheses + t_{obs})

TEST 3A 2025 · QUESTIONS 3 & 4 REFER TO THIS SCENARIO

A manufacturer **claims** its light bulbs last **1200 hours** on average. A sample of **25 bulbs** averages **1180 hours** with a **50-hour standard deviation**. Is there evidence that the true mean lifespan μ **differs** from the claimed 1200 hours? (Assume normal distribution.)

Q3: State the hypotheses H_0 vs H_1 and compute the **observed value of the test statistic** if H_0 is true.

 FORMULA CHOICE

1 sample, σ unknown, $n=25 < 30$ → **t-test**. **"Differs"** → **two-tailed**: $H_0: \mu=1200$, $H_1: \mu \neq 1200$. $df = n-1 = 24$.

[Sheet](#) → [S22: Case III \(\$\sigma\$ unknown, \$n < 30\$ → T\)](#)

Hypotheses. $H_0: \mu = 1200$ $H_1: \mu \neq 1200$

$t_{\text{obs.}}$ $t = (\bar{x} - \mu_0) / (s/\sqrt{n}) = (1180 - 1200) / (50/\sqrt{25}) = -20 / 10$ **$t_{\text{obs}} = -2$, $df = 24$**

2025 · Q4 Single Mean — Light Bulb Lifespan (p-value)

TEST 3A 2025 · Q4 (SAME SCENARIO AS Q3)

Compute the p-value for the hypothesis test in Q3. **FORMULA CHOICE**Two-tailed t-test, $df=24$. $p\text{-value} = 2 \cdot P(t_{24} > |t_{\text{obs}}|) = 2 \cdot P(t_{24} > 2)$. [Sheet → S22: Case III \(\$\sigma\$ unknown, \$n < 30 \rightarrow T\$ \)](#)**Step 1.** t-table: $t_{0.05,24}=1.711$, $t_{0.025,24}=2.064$. Since $1.711 < 2 < 2.064 \rightarrow 0.025 < P(t_{24} > 2) < 0.05$ **Step 2.** $p\text{-value} = 2 \times \text{that range} \rightarrow$ **$0.05 < p\text{-value} < 0.10$** At $\alpha=0.05$: $p\text{-value} > 0.05 \rightarrow$ Do not reject H_0 . Insufficient evidence that mean lifespan differs from 1200 hrs.**2025 · Q5 Two Proportions — Soldering Technique Defect Rate**

TEST 3A 2025 · Q5

A manufacturer compares two soldering techniques to see if the **new technique reduces defective boards**. Current technique: **56 defective out of 800**. New technique: **42 defective out of 900**. Conduct a hypothesis test to determine if the new technique results in a **lower proportion** of defective boards. Compute the observed value of the test statistic.

 FORMULA CHOICE

Two groups, proportion outcome $\rightarrow$ **two proportions z-test**. "**Lower proportion**" for new $\rightarrow H_1: p_1 - p_2 > 0$ (one-tailed upper, old minus new). Must use **pooled $\hat{p}_p$** for H-test. [Sheet → Two Proportions card → \$z_{\text{obs}}\$ row, pooled \$\hat{p}_p\$](#)

Step 1. $\hat{p}_1 = 56/800 = 0.07$ (current) $\hat{p}_2 = 42/900 = 0.0467$ (new)**Step 2.** $\hat{p}_p = (56+42)/(800+900) = 98/1700 = 0.0576$ **Step 3.** $z = (0.07 - 0.0467) / \sqrt{(0.0576 \times 0.9424 \times (1/800 + 1/900))} = 0.0233 / 0.01144$ **$z_{\text{obs}} = 2.04$** $p\text{-value} = P(Z > 2.04) \approx 0.0207 < 0.05 \rightarrow$ Reject H_0 . New technique has a significantly lower defect rate.

2025 · Q6 Paired t — Cooling System Temperature Reduction
TEST 3A 2025 · Q6

An engineer investigates the effect of a **new cooling system** on processor temperatures. The **same 6 processors** are measured before and after installing the system:

Processor	1	2	3	4	5	6
Before (°C)	85	90	88	92	87	89
After (°C)	78	84	80	85	82	83
d = Before–After	7	6	8	7	5	6

FORMULA CHOICE

"Same processors" measured under both conditions → **Paired t-test**. $n=6 < 30$. CI for μ_D .

[Sheet → S26: Case II \(n<30, Paired T, df=n-1\)](#)

Step 1. $\bar{d} = (7+6+8+7+5+6) / 6 = 39/6 = 6.5$

Step 2. $\Sigma d^2 = 49+36+64+49+25+36 = 259$ $s_d^2 = [259 - 6(6.5)^2] / 5 = [259 - 253.5] / 5 = 1.1$ $s_d = 1.0488$

Step 3. $df=5$, $t_{0.025,5} = 2.571$. CI: $6.5 \pm 2.571 \times (1.0488/\sqrt{6}) = 6.5 \pm 2.571 \times 0.4282 = 6.5 \pm 1.1008$

(5.3992, 7.6008) — We are 95% confident the new cooling system reduces mean temperature by between 5.40°C and 7.60°C.

2025 · Q7 Regression — Pipe Diameter vs Flow Rate (R^2)
TEST 3A 2025 · QUESTIONS 7–9 REFER TO THIS SCENARIO

A mechanical engineering team examined the relationship between **pipe diameter X** and **flow rate Y** (liters/min) across **n = 15 pipes**. Statistics: $\Sigma x_i = 300$, $\Sigma y_i = 4,500$, $\Sigma x_i y_i = 106,300$, $s_x = 10.5$, $s_y = 120.3$. Both X and Y are normally distributed.

Q7: Calculate the **coefficient of determination R^2** .

FORMULA CHOICE

Regression from summary stats. Compute S_{xy} first, then r, then R^2 .

[Sheet → Regression card → Sums block first](#)

Step 1. $\bar{x} = 300/15 = 20$ $\bar{y} = 4500/15 = 300$

Step 2. $S_{xy} = (1/(n-1)) [\Sigma x_i y_i - n \cdot \bar{x} \cdot \bar{y}] = (1/14) [106,300 - 15 \times 20 \times 300] = (1/14) [106,300 - 90,000] = 16,300/14 = 1164.29$

Step 3. $r = S_{xy} / (s_x \cdot s_y) = 1164.29 / (10.5 \times 120.3) = 1164.29 / 1263.15 = 0.9217$

Step 4. $R^2 = r^2 = 0.9217^2$ **$R^2 = 0.8496$** (84.96% of flow rate variation explained by pipe diameter)

2025 · Q8 Regression — Pipe Diameter vs Flow Rate (Line)

TEST 3A 2025 · Q8 (SAME SCENARIO AS Q7)

Determine the equation of the regression line. **FORMULA CHOICE**

$$b_1 = S_{xy} / s_{x^2} \cdot b_0 = \bar{y} - b_1 \bar{x}$$
[Sheet → Regression card → \$b_1, b_0\$ formulas](#)

$$b_1: b_1 = S_{xy} / s_{x^2} = 1164.29 / 10.5^2 = 1164.29 / 110.25 \quad \mathbf{b_1 = 10.56}$$

$$b_0: b_0 = \bar{y} - b_1 \bar{x} = 300 - 10.56 \times 20 = 300 - 211.2 \quad \mathbf{b_0 = 88.8}$$

Line. Regression equation: $\mathbf{\hat{y} = 88.8 + 10.56 X}$

2025 · Q9 Regression — Test for Linear Relationship (p-value)

TEST 3A 2025 · Q9 (SAME SCENARIO AS Q7)

Test the hypotheses $H_0: \rho = 0$ vs $H_a: \rho \neq 0$ for a linear relationship between X and Y. Compute the **p-value**. **FORMULA CHOICE**

Test using r. $t = r \sqrt{(n-2)/(1-r^2)} \sim t_{\{n-2\}}$. Two-tailed since $\rho \neq 0$. [Sheet → Regression card → correlation test](#)

Step 1. $t = r \times \sqrt{(n-2)/(1-r^2)} = 0.9217 \times \sqrt{13/(1-0.8496)} = 0.9217 \times \sqrt{13/0.1504} = 0.9217 \times 9.298 \quad \mathbf{t_{obs} = 8.567, df = 13}$

Step 2. Two-tailed: $p\text{-value} = 2 \cdot P(t_{13} > 8.567)$. $t_{0.0005, 13} \approx 4.221 < 8.567 \rightarrow \mathbf{p\text{-value} < 0.001 (\approx 0)}$

Reject H_0 . Strong evidence of a linear relationship between pipe diameter and flow rate.

2025 · Q10 Two Independent Means — Battery Life (Full Answer)
TEST 3A 2025 · Q10 (FULL ANSWER — 6 MARKS)

An experiment compared the battery life of **Lithium-ion** vs **NiMH** batteries:

Lithium-ion: $\bar{x}_1 = 450$ hrs, $s_1 = 21$ hrs, $n_1 = 20$. NiMH: $\bar{x}_2 = 400$ hrs, $s_2 = 25$ hrs, $n_2 = 15$.

Is there evidence that **Lithium-ion battery life exceeds NiMH**? Assume normal distributions. Let $\mu_1 - \mu_2$ = difference in population means.

(a) State H_0 and H_1 . (b) State the test statistic and its distribution. (c) Find t_{obs} . (d) Compute/bracket the p-value. (e) Reject H_0 at $\alpha=0.05$?

FORMULA CHOICE

$s_2/s_1 = 25/21 = 1.19 \leq 1.4 \rightarrow$ **Pooled T**. Small samples ($n_1=20, n_2=15$ both < 30). **"Exceeds"** $\rightarrow$ one-tailed upper.

[Sheet \$\rightarrow\$ S24/25: Case III \(Pooled T, small n, ratio \$\leq 1.4\$ \)](#)

(a) Hypotheses

$$H_0: \mu_1 - \mu_2 \leq 0 \quad H_1: \mu_1 - \mu_2 > 0$$

(b) Test statistic & distribution

$$\text{Ratio} = 1.19 \leq 1.4 \rightarrow T = [(\bar{x}_1 - \bar{x}_2) - 0] / [s_p \sqrt{(1/n_1 + 1/n_2)}] \sim t_{\{n_1+n_2-2\}} = t_{33}$$

(c) t_{obs}

$$s_p^2: s_p^2 = [(n_1-1)s_1^2 + (n_2-1)s_2^2] / (n_1+n_2-2) = [19(441) + 14(625)] / 33 = [8379+8750] / 33 = 17129/33 = 519.06 \quad s_p = 22.783$$

$$\text{ese. } \text{ese} = 22.783 \times \sqrt{(1/20 + 1/15)} = 22.783 \times 0.3416 = 7.782$$

$$t. \quad t = (450 - 400) / 7.782 = 50 / 7.782 \quad \mathbf{t_{\text{obs}} = 6.4252, df = 33}$$

(d) p-value

$$\text{One-tailed upper: } p\text{-value} = P(t_{33} > 6.4252). t_{0.0005,33} \approx 3.611 << 6.43 \rightarrow \mathbf{p\text{-value} < 0.0005}$$

(e) Conclusion at $\alpha = 0.05$

$p\text{-value} < 0.0005 < \alpha = 0.05 \rightarrow$ **Reject H_0 .**

Strong evidence that Lithium-ion battery life exceeds NiMH battery life on average.